The Short-Time Fourier Transform and the Spectrogram

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Key Concepts

1) The short-time Fourier transform (STFT) is used for analysis of signals with time-varying spectral characteristics.

2) The STFT is based on applying the DFT to windowed segments of the signal.

    a) The segment captures the characteristics in the corresponding time interval.

    b) The DFT coefficients are displayed as a function of time and frequency, where time refers to the segment location.

3) The spectrogram is the magnitude squared of the STFT coefficients.

4) The temporal window length L determines both temporal and frequency resolution.

    a) Shorter window lengths L allow more rapid tracking of changes in the spectral characteristics over time.

    b) Longer window lengths L provide better frequency resolution - increased ability to see closely-spaced details in the spectral characteristics.

5) The same tradeoffs between dynamic range and resolution that apply to the DFT when choosing a window also apply to the STFT.

6) The spectrogram can be viewed as the power at the output of a bank of filters.

    a) Each filter has the same frequency response, but shifted to a different center frequency.

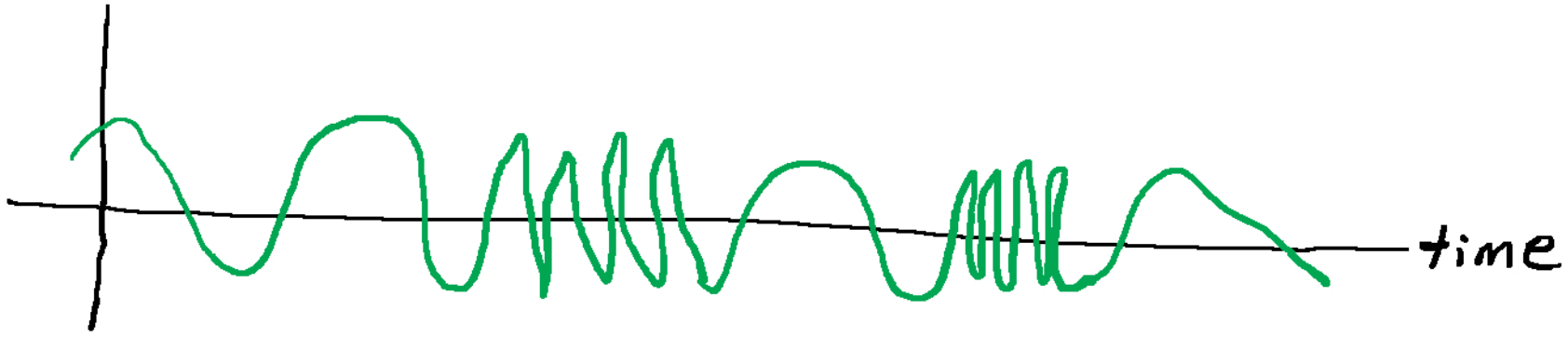
    b) The frequency response of the filters are determined by the window.

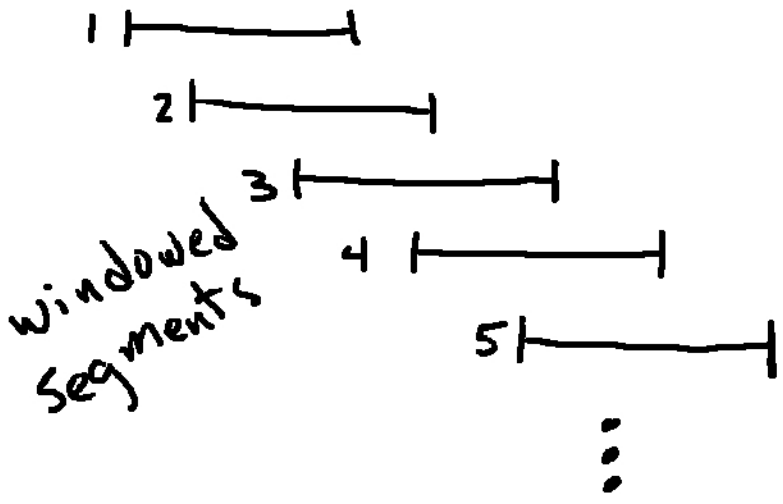
Video Screenshots

Analysis of time-varying spectral characteristics

0:10

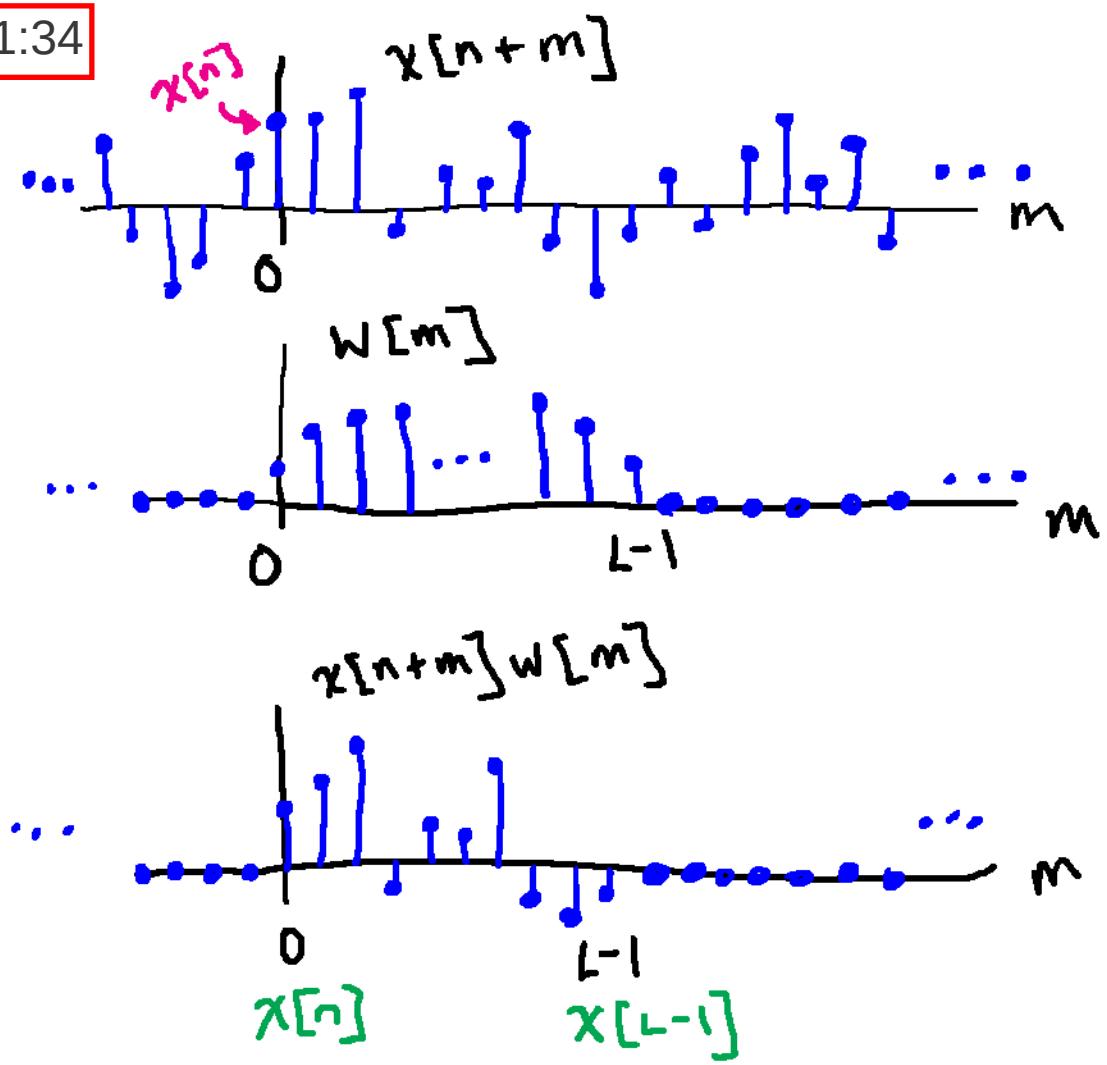
- speech
- music
- seismology
- ⋮





Apply DFT to windowed segments of data

Display DFT coefficients as a function of time and frequency



$$X[n,\lambda] = \sum_{m=-\infty}^{\infty} x[n+m]\,w[m]\,e^{-j\lambda m}$$

"location in time" — n; frequency — λ; DTFT of $x[n+m]w[m]$

Sample λ at $\lambda_k = \frac{2\pi}{N}k$

- length $N > L$ DFT

$$X[n,k] = \sum_{m=0}^{L-1} x[n+m]\,w[m]\,e^{-j k\frac{2\pi}{N} m}$$

STFT

involves $x[n], x[n+1], \ldots, x[n+L-1]$

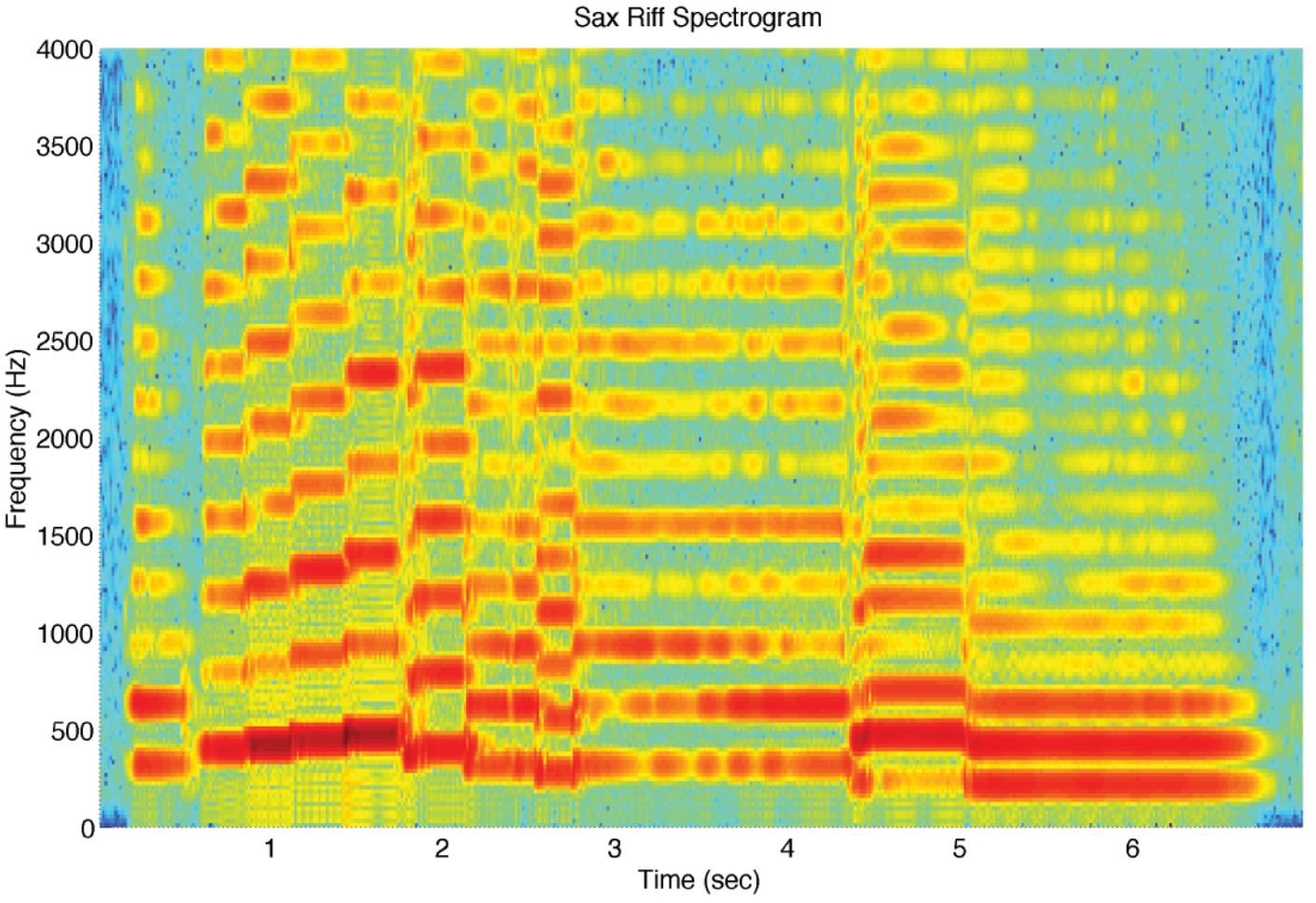
Spectrogram: $|X[n,k]|^2$

Inverse STFT:

$$x[n+m] = \frac{1}{N w[m]} \sum_{k=0}^{N-1} X[n,k]\, e^{j\frac{2\pi}{N}km}$$

Saxophone Riff

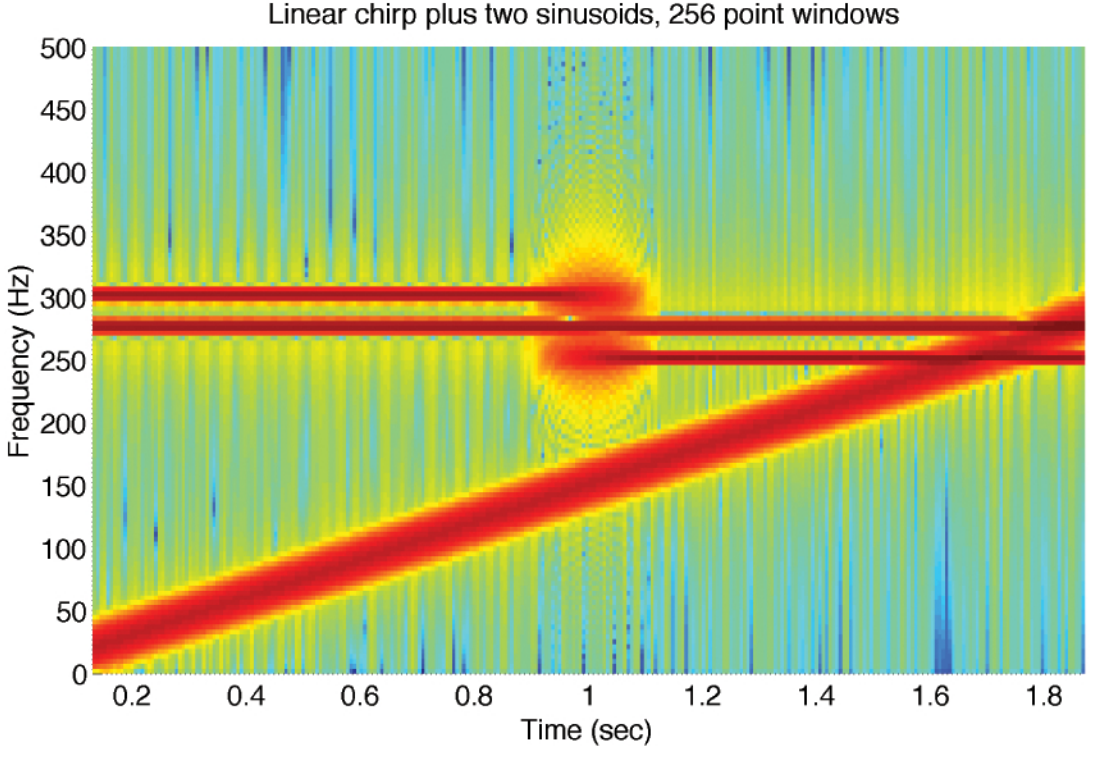
- $f_s = 44{,}100$ kHz
- $L = 812$ pts/segment
- $N = 2048$ freq. samples
- 192 point overlap



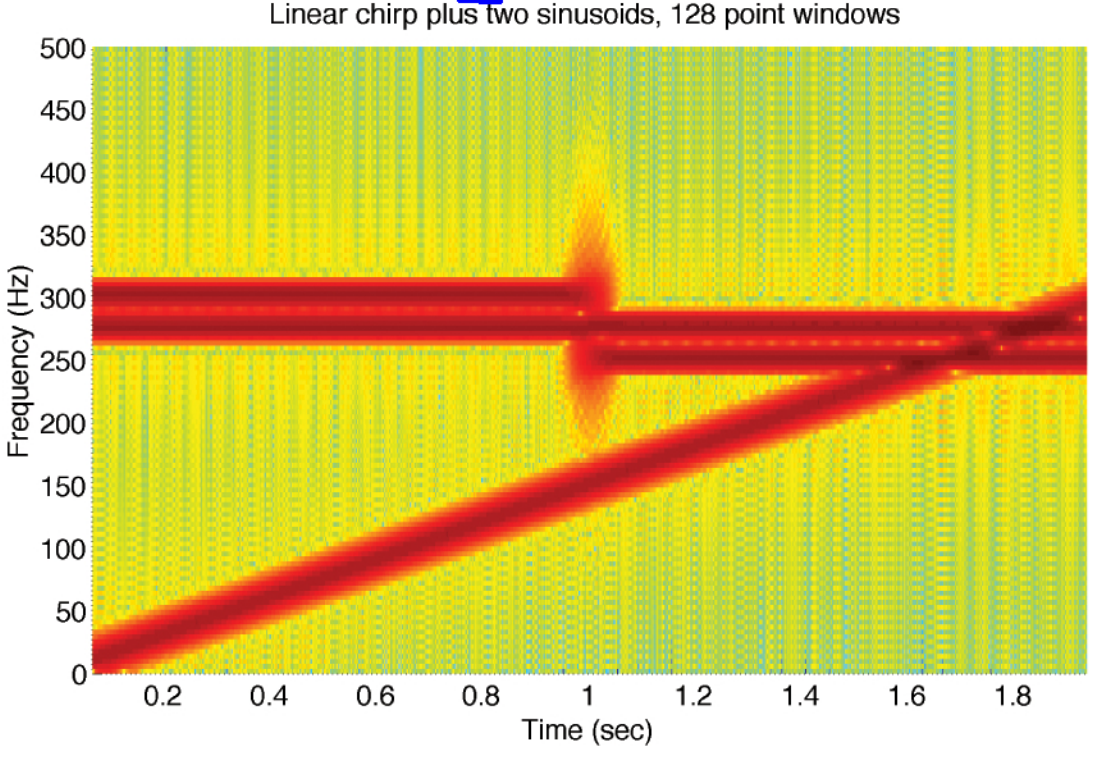
L trades temporal resolution for frequency resolution

Example: linear chirp 0-300 Hz; sin #1 : 275 Hz;
sin #2: 300 Hz 0-1s, 250 Hz 1-2s; f_s = 1000 Hz

L = 256

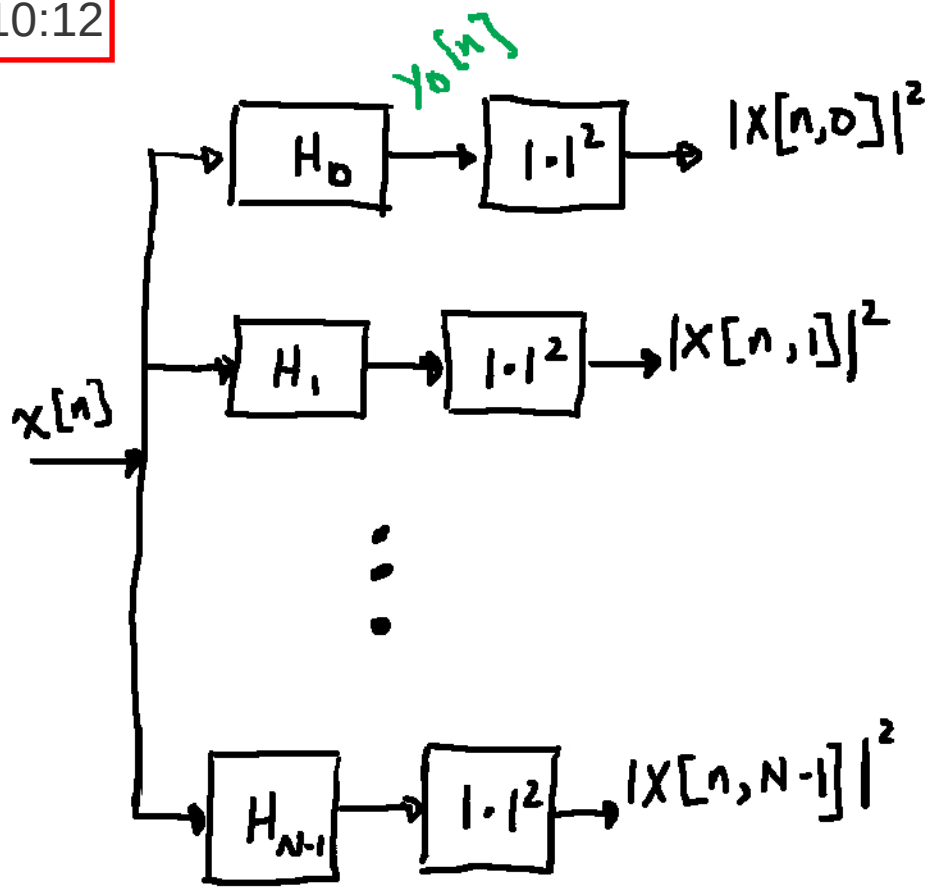


L = 128

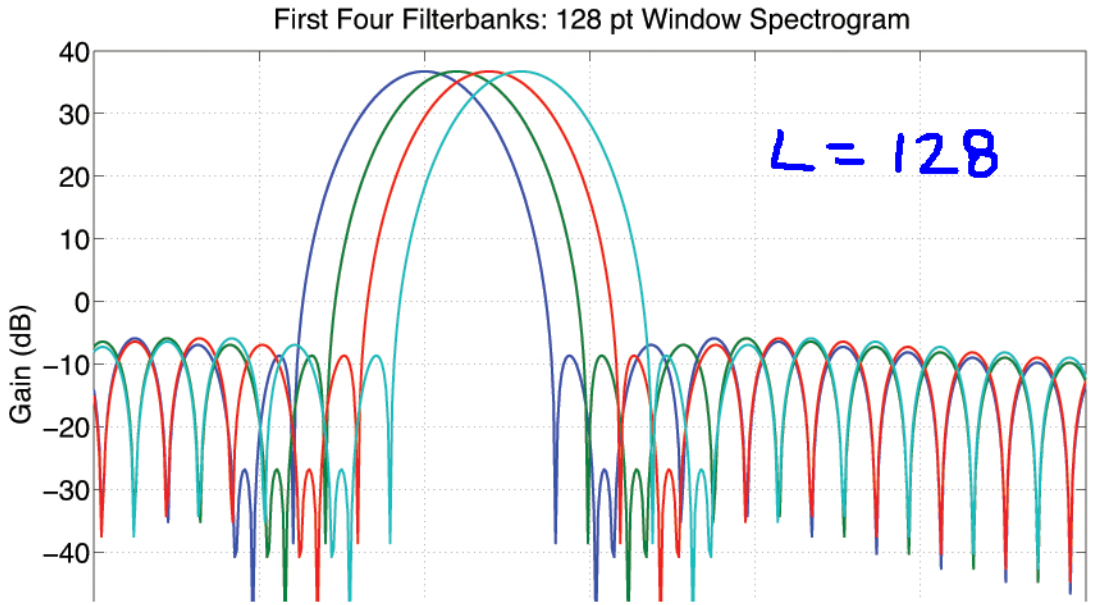


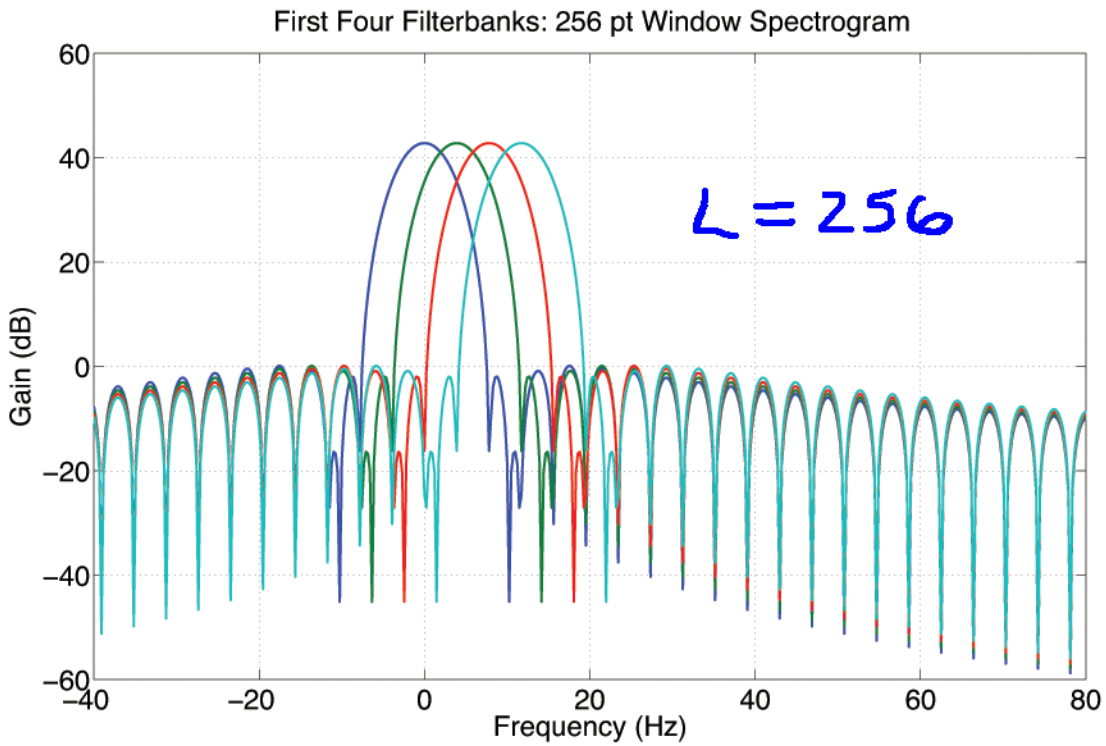
Filter Bank Interpretation

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L increases: bandwidth decreases and impulse response "duration" increases





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